

- Points pre-assigned to review questions and problems are just an indication of the final score: partial and unsatisfactory answer/solution will receive a fraction of the points, whereas excellent answers may exceed the indicated points.
- Use of books/notes is forbidden. A list of useful formulas and equations is provided below. Calculators are not needed to solve problems.
- Indicate name and UC ID number in ALL the solution sheets. Number the solution sheets and indicate total number of sheets (e.g., first sheet 1/3, second sheet 2/3, third sheet 3/3).

Equations you MAY need:

Exponential distribution (rate λ , τ : interarrival time)

$$P(\tau > t) = e^{-\lambda t}, \quad \text{pdf: } f_x(x) = \lambda e^{-\lambda x}. \quad (1)$$

M/M/1 system, infinite buffer (arrival rate λ , service rate μ , $\rho = \lambda/\mu$):

Probability that the buffer is empty:

$$\pi(0) = \lim_{t \rightarrow \infty} P(N(t)=0) = 1 - \rho. \quad (2)$$

Probability that n packets are in the system

$$\pi(n) = \lim_{t \rightarrow \infty} P(N(t)=n) = \rho^n \pi(0). \quad (3)$$

Average time in the system and average number of packets in the system(infinite buffer):

$$E[T] = \frac{1/\mu}{1-\rho} \quad E[N] = \frac{\rho}{1-\rho} \quad (4)$$

M/M/1/K, finite buffer of size K :(arrival rate λ , service rate μ , $\rho = \lambda/\mu$):

Probability that the buffer is empty:

$$\pi(0) = \lim_{t \rightarrow \infty} P(N(t)=0) = \frac{1-\rho}{1-\rho^{K+1}}. \quad (5)$$

Probability that n packets are in the system

$$\pi(n) = \lim_{t \rightarrow \infty} P(N(t)=n) = \rho^n \pi(0). \quad (6)$$

Average time in the system and average number of packets in the system(infinite buffer):

$$E[T] = \frac{E[N]}{\lambda(1-\pi(K))} \quad E[N] = \frac{\rho}{1-\rho} - \frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}} \quad (7)$$

Erlang B formula (M/M/c/c system, arrival rate λ , duration μ , $a = \lambda/\mu$)

$$\lim_{t \rightarrow \infty} P(N(t)=0) = \frac{1}{\sum_{k=0}^c a^k/k!} \quad \lim_{t \rightarrow \infty} P(N(t)=n) = \frac{a^n/n!}{\sum_{k=0}^c a^k/k!} \quad (8)$$

Series $0 < p < 1$

$$\sum_{n=1}^{\infty} n p^{n-1} = p/(1-p)^2 \quad (9)$$

—— Review questions 1

R1.a (4Points) Describe the slotted Aloha protocol. Does the protocol provide adaptation to congestion? Explain your answer.

R1.b (5Points) Describe multiplexing and demultiplexing of TCP and UDP protocols.

R1.c (5Points) Discuss why we need hierarchical routing in the internet. (Very) Briefly describe intra-AS and inter-AS routing.

R1.d (6Points) Compare and contrast the Link State and the Distance Vector algorithm.

—— Review questions 2

R2.a (7Points) List and briefly describe the components of a router and list the types of switching fabric. Do the input and output ports implement protocols? What layer do they refer to?

R2.b (7Points) List the main examples of "taking turns" and "Random access" Medium Access Control (MAC) protocols and briefly describe them. Discuss their advantages/disadvantages. In what traffic regimes do they best perform? Explain your answer.

R2.c (8Points) Provide a brief overview of TCP congestion control. Why the timeout event and the duplicate acknowledgment events generate a different reaction of the protocol?

R2.d (8Points) Describe uncontrolled flooding, controlled flooding and spanning tree broadcast algorithms. Rank these techniques according to the number of packet replicas they generate. Which of them require a preliminary route set-up phase?

Problems 1

3

P1.a (4Points) Consider a network with 3 terminals (A, B and C). Terminals implement slotted aloha, and have transmission probability equal to $p=0.2$. The size of packets is $L=0.5$ Kbits, and the duration of one slot is $T=1$ ms. The transmission of one packet fits the duration of a slot.

- Find the transmission rate of the terminals.
- Find the average throughput of terminal A.
- Find the average throughput of the network.

P1.b (5Points) Consider a router with two input ports (input port 1 and 2) and two output ports (output port 1 and 2). The transfer rate of a packet from an input port to an output port is $R_{in}=1\text{Mbps}$ (deterministic transfer rate). At time $t=0$, packet A just arrived in the buffer of the input port 1 and the buffer of the input port 2 is empty. Packet A needs to be transferred to output port 1. At time $t=0.05$ s, packet B arrives in the buffer of the input port 2. Packet B needs to be transferred to the output port 2. The size of packet A and B is $L=100\text{Kbit}$.

- Assume the switching fabric of the router is a bus. Compute the time at which the last bit of B is transferred to the output port 2.
- Compute the same quantity assuming the switching fabric of the router is a crossbar.
- Assume the transfer rate of the output ports (from the output port to the outside of the router) is equal to $R_{out}=0.5\text{Mbps}$. Compute the time at which the last bit of B is transmitted from the output port 2 in both cases.

$$\frac{100\text{K}}{1\text{M}} = 0.1$$

P1.c (5Points) Consider the 4 node topology in Fig. 1. The weight on the links ($C(n_1, n_2)$, positive integer) represent the cost of link.

- Assume distance vector is used, show the routing tables of the nodes at the various iterations.
- Find the maximum cost $C(A, B)$ such that node A advertises a new route to node D. Find the minimum cost $C(A, C)$ such that node A advertises a new route to node B (use the original costs).

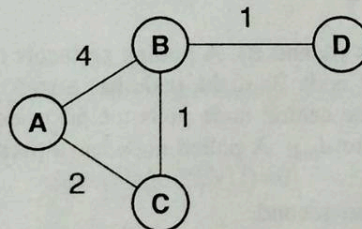
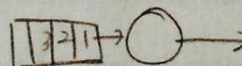


Fig. 1. Problem P1.c

P1.d (6Points) Consider a M/M/1/5 server (5 is the size of the buffer in packets) with packet arrival rate equal to $\lambda=1$ packet/s and packet service rate equal to $\mu=2$ packet/s.

- Find the probability that a packet arriving in the buffer is dropped by the server.
- Find the probability that the server is idle (that is, it is not serving any packet).
- 3 packets are in the buffer. Find the average time the packets spend in the buffer (include the service time). What is the time the 3rd packet spends in the buffer?
- Assume 1 packet is in the buffer and is being served. What is the probability that the next packet arrives in the buffer before the service terminates?



Problems 2

P2.a (7Points) Consider the topology in Fig. 2, where the rate of link 1 and 2 is R_1 bps and R_2 , respectively. The source S transmits packets of size L bits to destinations D_1 , D_2 and D_3 . Connections sharing a link are

assigned equal rate. In the unicast case, the source creates replicas and sequentially injects them according to the destination index. Assume that in the multicast case a link connected to more than one destination sequentially transmits packets according to the index of the destinations (that is, the link creates the replicas and sequentially transmit them).

- Assume the source uses unicast connections to the destinations. Find the throughput to D_1 .
- Assume the source uses multicast connections to the destinations. Find the throughput to D_1 .
- Assume the source injects in link 1 packet A and B back to back. Find the inter arrival delay in the two cases (unicast and multicast).
- Assume $R_1 = R_2$, find the overall delay at destination D_3 (time to reception of the last bit of packet B) in the unicast and multicast case.

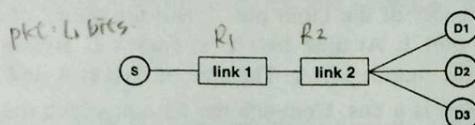


Fig. 2. Problem P2.a

P2.b (7Points) A server accepts at most 3 TCP connections (i.e., at most 3 connections can be active simultaneously). The duration of a connection is a random variable distributed according to an exponential distribution with rate $\mu=1$ connection/min. Connection request interarrivals is a random variable distributed according to an exponential distribution with rate $\lambda=2$ connection/min.

- Compute the probability that an incoming request is not accepted.
- Assume two connections are active, compute the probability that the next request arrives before one of the two connections terminates.
- Assume one connection is active. Find the probability that at least 2 new connection requests arrive before that connection terminates.

P2.c (8Points) Consider two wireless nodes (A and B). A polling protocols is used to regulate channel access. A central node sequentially polls node A and node B: if the node has a packet to transmit then it is assigned the channel for a time equal to T , otherwise the central node polls the next node. The transmission rate is equal to R bps and the time to poll a node is equal to d_{poll} . A polled node has a packet to transmit with probability equal to p .

- Find the throughput of node A in bit per second.
- Find the overall throughput of the network in bit per second.
- Find the throughput of node A assuming now that the probability that node A has a packet to transmit when polled is equal to $2p$ (the probability that node B has a packet to transmit when polled is equal to p). Is the throughput of node A twice the throughput computed when the transmission probability is equal to p ?

P2.d (8Points) Consider a router with two input ports (input port 1 and 2) and two output ports (output port 1 and 2). The transfer time from any input port to any output port is a random variable distributed according to an exponential distribution with average time $1/\mu$. The transmission time from any output port to the outside is a random variable distributed according to an exponential distribution with average time $1/\mu$. At time $t=0$, packet A is being transferred to the output port 1. Packet B arrives at time τ in the input port 2 and needs to be transferred to the output port 2. The time τ is a random variable distributed according to an exponential distribution with average time $1/(2\mu)$ (no more packets arrive).

- Compute the average time B spends in the buffer of the input port 2 (include the transfer time) in the case the switching fabric is a bus. Compute the same quantity assuming now that the switching fabric is a crossbar.
- Compute the expectation of T_A , where T_A is the time when the last bit of packet A leaves the router (bus and crossbar cases).
- Compute the expectation of T , where T is the time when both the packets are out of the router (bus case).